

(12) **United States Patent**
Chang

(10) **Patent No.:** **US 12,389,592 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **MEMORY DEVICE AND METHOD OF FORMING THE SAME**

(71) Applicant: **NANYA TECHNOLOGY CORPORATION**, New Taipei (TW)

(72) Inventor: **Chin-Piao Chang**, New Taipei (TW)

(73) Assignee: **NANYA TECHNOLOGY CORPORATION**, New Taipei (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 617 days.

(21) Appl. No.: **17/810,611**

(22) Filed: **Jul. 4, 2022**

(65) **Prior Publication Data**

US 2024/0008269 A1 Jan. 4, 2024

(51) **Int. Cl.**
H10B 20/20 (2023.01)
G11C 17/16 (2006.01)
H01L 21/283 (2006.01)
H01L 21/311 (2006.01)
H01L 21/768 (2006.01)
H01L 23/525 (2006.01)

(52) **U.S. Cl.**
CPC **H10B 20/20** (2023.02); **G11C 17/16** (2013.01); **H01L 21/283** (2013.01); **H01L 21/76877** (2013.01); **H01L 23/5252** (2013.01); **H01L 21/31105** (2013.01); **H01L 2221/101** (2013.01); **H01L 2221/1068** (2013.01)

(58) **Field of Classification Search**
CPC H01L 23/5252; H01L 23/5254; H01L

2221/101–1015; H01L 21/31105–31122; H01L 21/311–31144; H01L 21/282–2885; H01L 21/44–449; H01L 21/76877–76883; H01L 21/47635; H01L 21/461; H01L 21/4763; H01L 21/76–765; H10D 84/0151; H10D 84/0188; H10D 30/795

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2008/0073755 A1* 3/2008 Ang H01L 23/5252
257/656
2014/0024210 A1* 1/2014 Yang H01L 21/7684
257/E21.586
2020/0212049 A1* 7/2020 Lin H10B 12/34
2020/0343182 A1* 10/2020 Lin H10B 20/25
2020/0343253 A1* 10/2020 Lin H10B 12/31
2023/0024465 A1* 1/2023 Li H10B 12/36

* cited by examiner

Primary Examiner — Zandra V Smith

Assistant Examiner — Douglas Yap

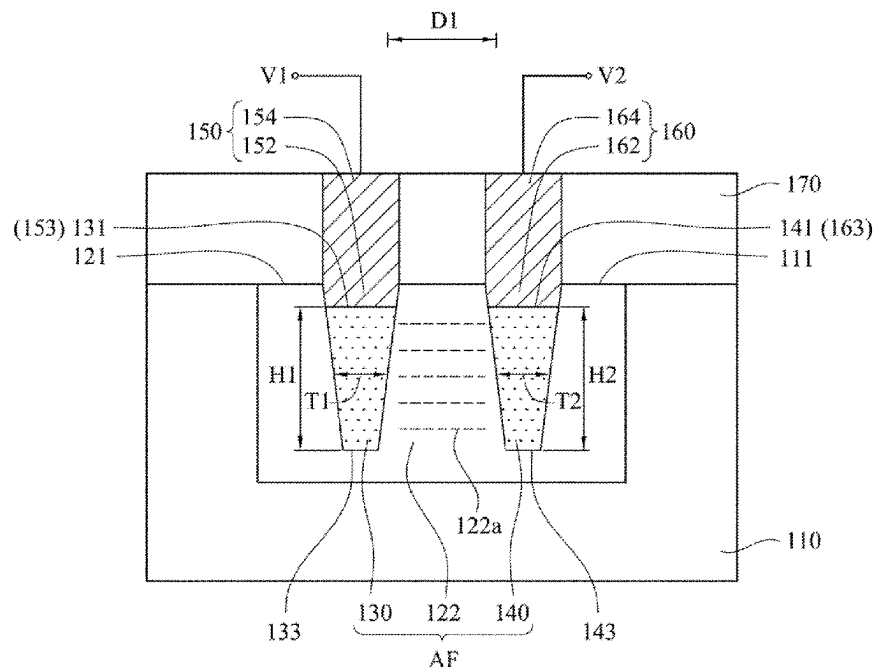
(74) *Attorney, Agent, or Firm* — CKC & Partners Co., LLC

(57) **ABSTRACT**

A memory device includes a semiconductor substrate, an isolation structure, and an anti-fuse structure. The isolation structure is disposed in the semiconductor substrate. The anti-fuse structure is disposed in the isolation structure and includes a first electrode and a second electrode. The second electrode is disposed adjacent to the first electrode. Both of a top surface of the first electrode and a top surface of the second electrode are below a top surface of the semiconductor substrate.

19 Claims, 6 Drawing Sheets

10



10

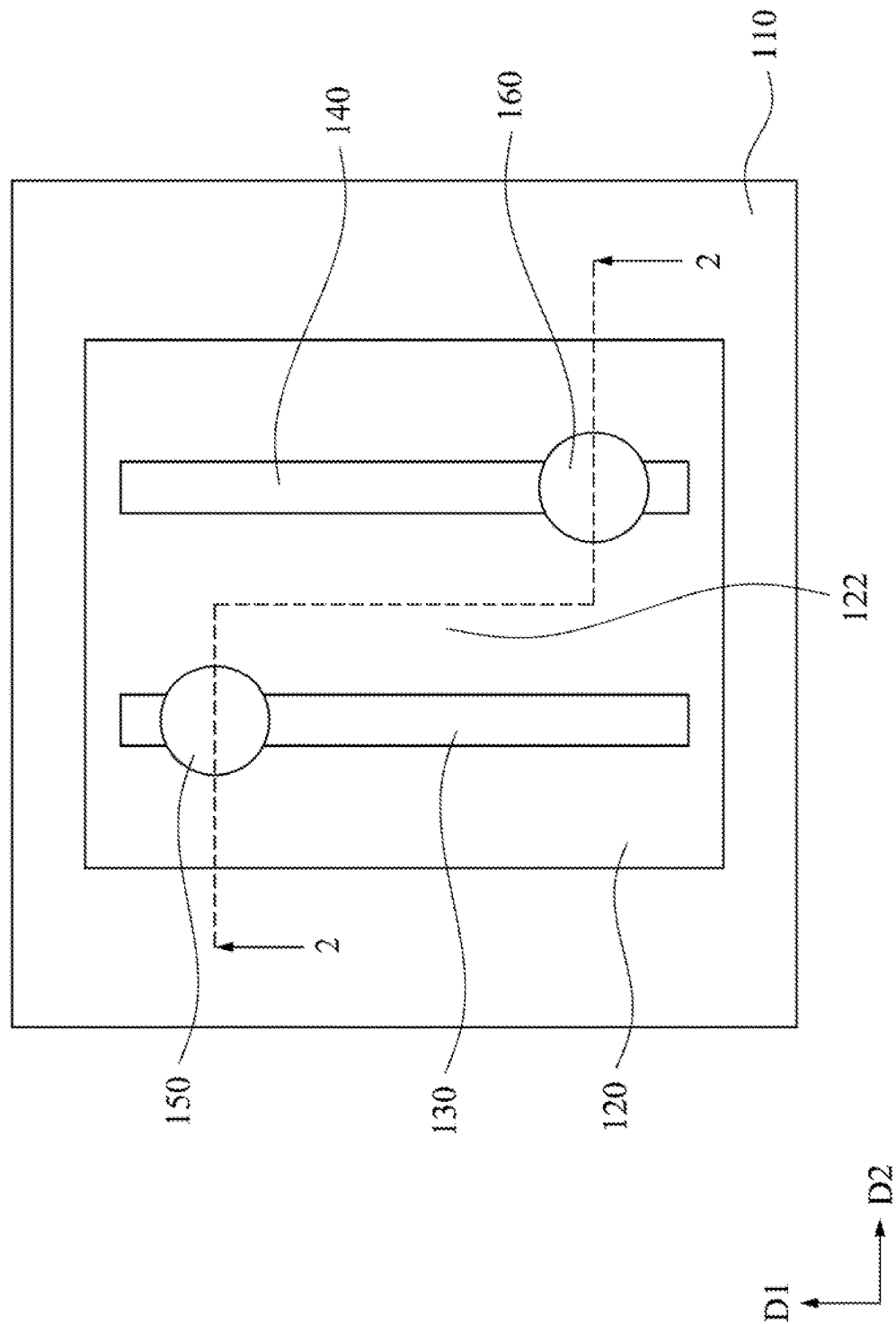


Fig. 1

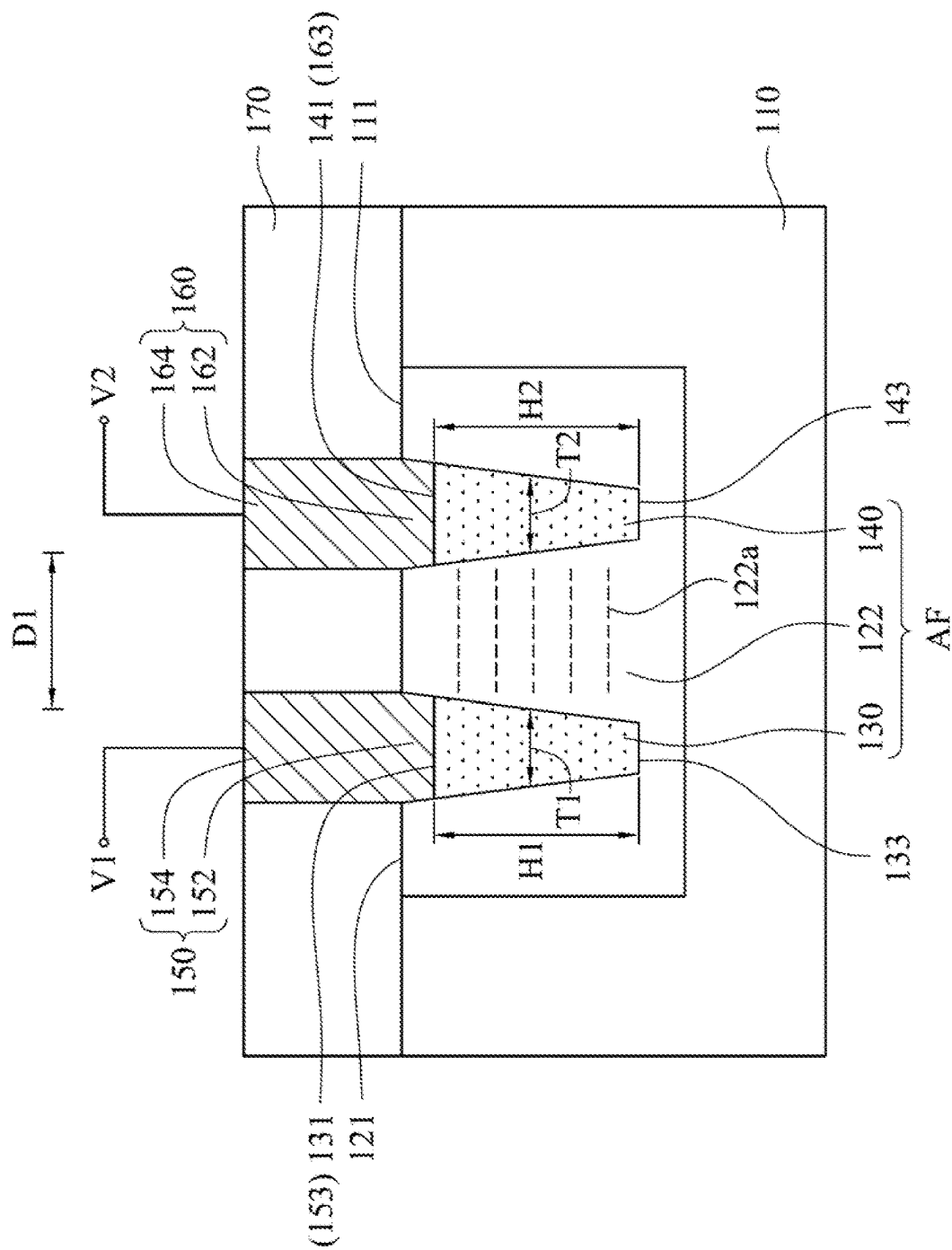


Fig. 2

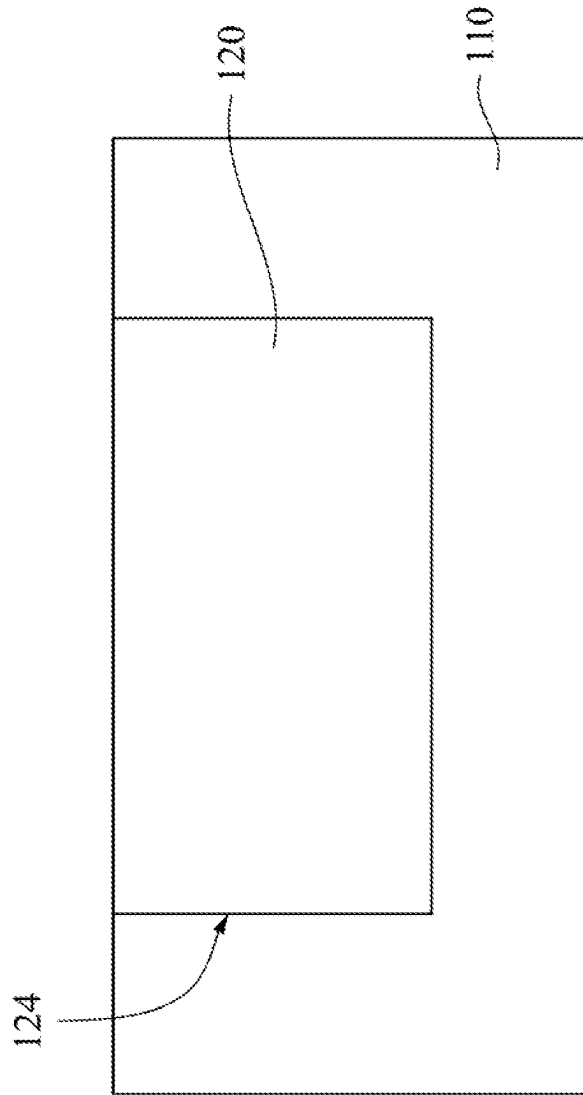


Fig. 3

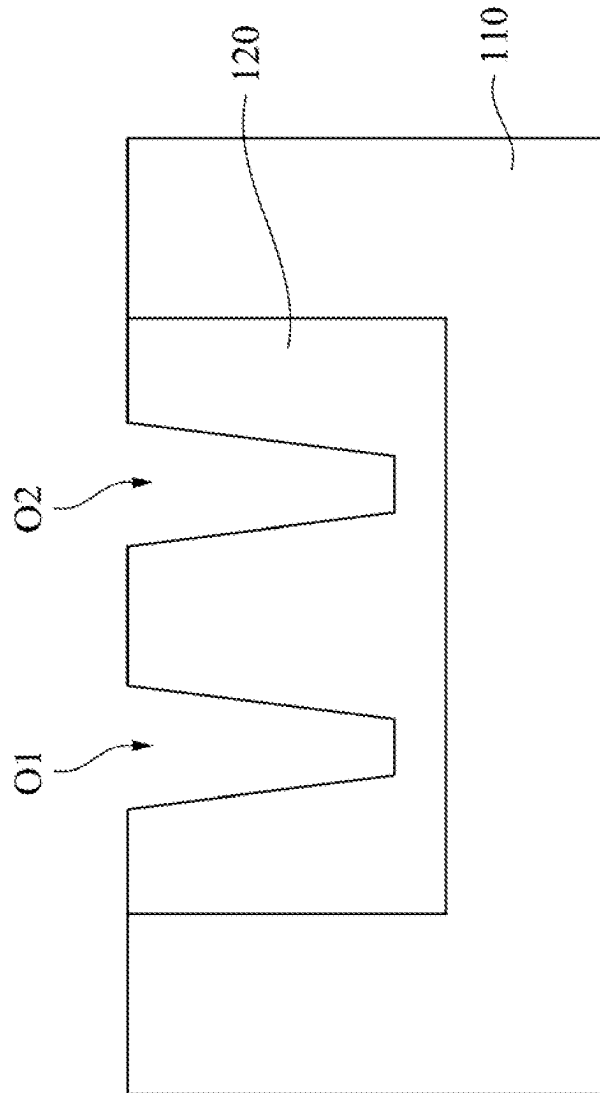


Fig. 4

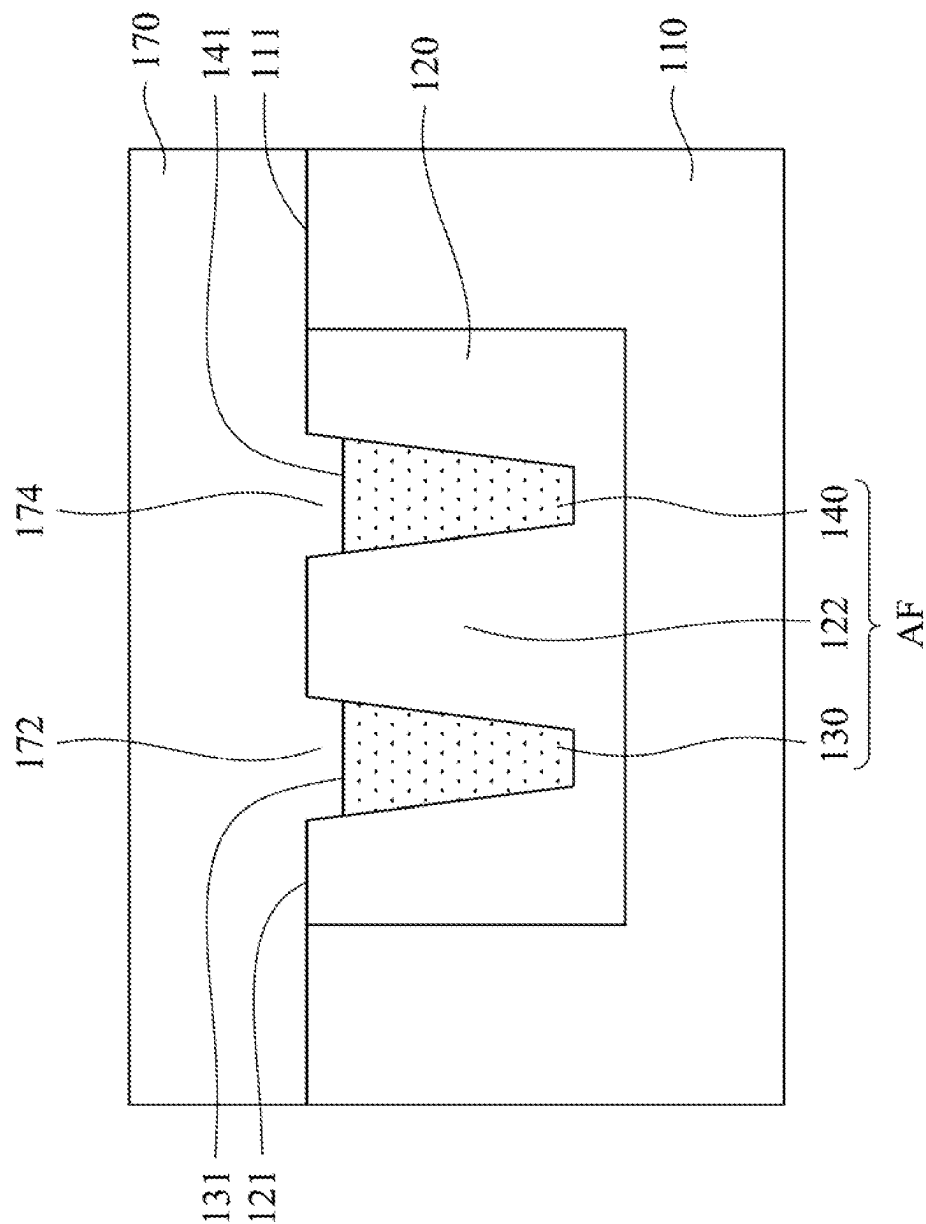


Fig. 5

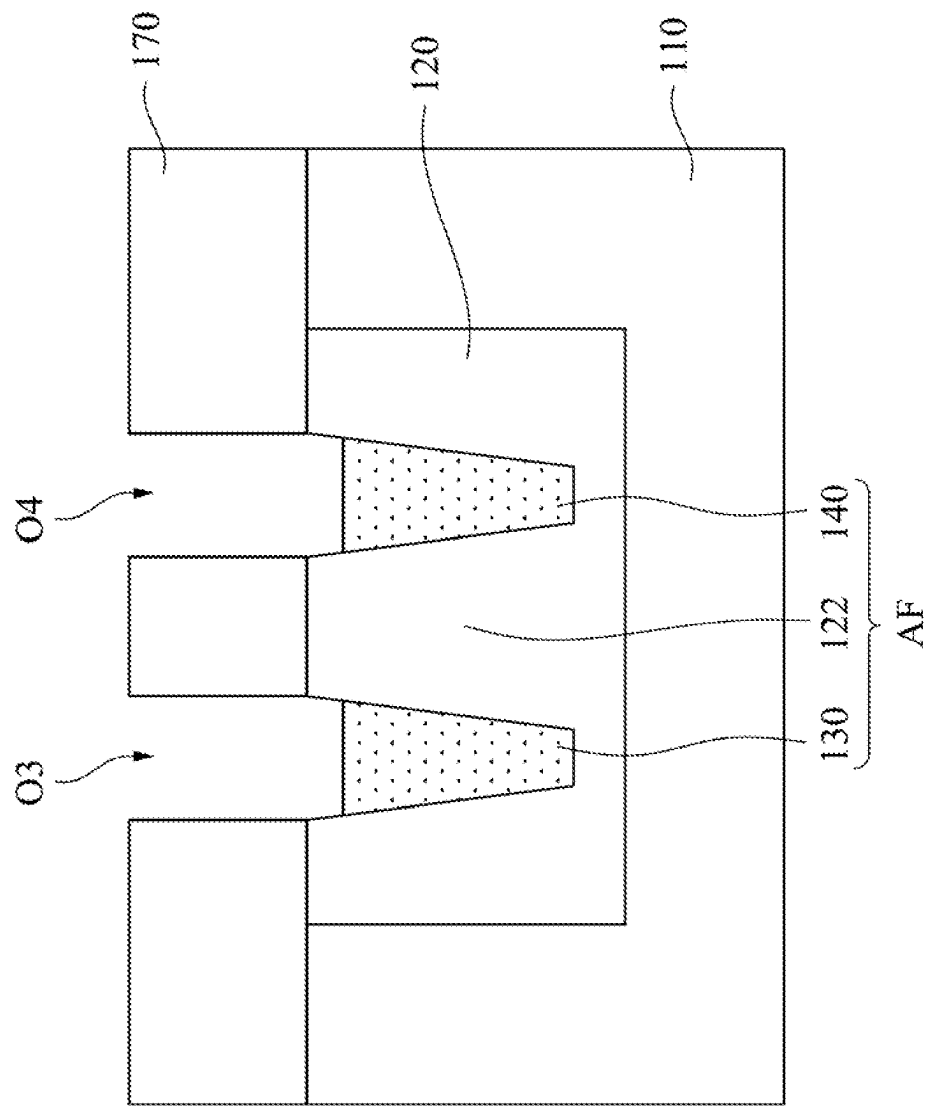


Fig. 6

1

MEMORY DEVICE AND METHOD OF FORMING THE SAME

BACKGROUND

Technical Field

The present disclosure relates to a memory device and a method of forming a memory device.

Description of Related Art

Semiconductor memory devices may be classified into two categories, volatile memory devices and nonvolatile memory devices. In contrast to the volatile memory devices, the information of the nonvolatile memory devices is preserved even with the power removed. In regards to the nonvolatile memory devices, some designs allow multiple programming, while other designs allow one-time programming. Typically, the manufacturing techniques used to form nonvolatile memory devices are quite different from a standard logic process, which dramatically increases the complexity and chip size.

SUMMARY

One aspect of the present disclosure is a memory device.

According to some embodiments of the present disclosure, a memory device includes a semiconductor substrate, an isolation structure, and an anti-fuse structure. The isolation structure is disposed in the semiconductor substrate. The anti-fuse structure is disposed in the isolation structure and includes a first electrode and a second electrode. The second electrode is disposed adjacent to the first electrode. Both of a top surface of the first electrode and a top surface of the second electrode are below a top surface of the semiconductor substrate.

In some embodiments, the memory device further includes a dielectric layer above the semiconductor substrate and the isolation structure.

In some embodiments, the first contact includes a bottom portion in the isolation structure and a top portion in the dielectric layer.

In some embodiments, the first contact is configured to apply a first voltage to the first electrode and the second contact is configured to apply a second voltage different from the first voltage to the second electrode to convert a portion of the isolation structure between the first electrode and the second electrode into a permanent electrically conductive path.

In some embodiments, both of the top surface of the first electrode and the top surface of the second electrode are below a top surface of the isolation structure.

In some embodiments, the top surface of the first electrode is substantially coplanar with the top surface of the second electrode.

In some embodiments, a bottom surface of the first electrode is substantially coplanar with a bottom surface of the second electrode.

In some embodiments, the memory device further includes a first contact and a second contact. The first contact is disposed on the first electrode. The second contact is disposed on the second electrode.

In some embodiments, both of a bottom surface of the first contact and a bottom surface of the second contact are below the top surface of the semiconductor substrate.

2

In some embodiments, the first electrode extends along a first direction, and the first electrode and the second electrode are arranged parallel with each other along a second direction perpendicular to the first direction in a top view.

In some embodiments, the first electrode has a strip profile in a top view.

In some embodiments, the first electrode and the second electrode include the same materials.

Another aspect of the present disclosure is a method of forming memory device.

According to some embodiments of the present disclosure, a method of forming memory device includes forming an isolation structure in a semiconductor substrate. The isolation structure is etched to form a first opening and a second opening. A first electrode and a second electrode of an anti-fuse structure are formed respectively in the first opening and the second opening such that both of a top surface of the first electrode and a top surface of the second electrode are below a top surface of the isolation structure. A dielectric layer is formed above the isolation structure. A first voltage is applied to the first electrode and a second voltage different from the first voltage is applied to the second electrode to convert a portion of the isolation structure between the first electrode and the second electrode into a permanent electrically conductive path.

In some embodiments, the method further includes forming a first contact and a second contact in the dielectric layer such that the first voltage is applied to the first electrode through the first contact and the second voltage is applied to the second electrode through the second contact.

In some embodiments, forming the first contact and the second contact is performed such that a portion of the first contact and a portion of the second contact are formed within the isolation structure.

In some embodiments, forming the first contact and the second contact is performed such that the top surface of the first electrode and the top surface of the second electrode are respectively in contact with the first contact and the second contact.

In some embodiments, forming the first electrode and the second electrode is performed further such that the top surface of the first electrode and the top surface of the second electrode are below a top surface of the semiconductor substrate.

In some embodiments, forming the first electrode and the second electrode is performed such that the first electrode and the second electrode are in contact with the isolation structure.

In some embodiments, forming the first electrode and the second electrode is performed such that a bottom surface of the first electrode is coplanar with a bottom surface of the second electrode.

In some embodiments, forming the isolation structure in the semiconductor substrate includes forming a trench in the semiconductor substrate and filling an anti-fuse dielectric material in the trench to form the isolation structure.

In the aforementioned embodiments, since both of the top surface of the first electrode and the top surface of the second electrode are below the top surface of the semiconductor substrate, a size of the memory device can be decreased. Further, a manufacturing process of the memory device can be simplified.

It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a top view of a layout of a memory device in accordance with some embodiments of the present disclosure.

FIG. 2 is a cross-sectional view of the memory device taken along line A-A of FIG. 1.

FIG. 3 to FIG. 6 are cross-sectional views of a method of forming the memory device at various stages in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

As used herein, “around,” “about,” “approximately,” or “substantially” shall generally mean within 20 percent, or within 10 percent, or within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around,” “about,” “approximately,” or “substantially” can be inferred if not expressly stated.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

FIG. 1 is a top view of a layout of a memory device 10 in accordance with some embodiments of the present disclosure, and FIG. 2 is a cross-sectional view of the memory device 10 taken along line 2-2 of FIG. 1. Referring to FIG. 1 and FIG. 2, a memory device 10 includes a semiconductor substrate 110, an isolation structure 120, and an anti-fuse structure AF. The isolation structure 120 is disposed in the semiconductor substrate 110. The anti-fuse structure AF is disposed in the isolation structure 120. The anti-fuse structure AF includes a first electrode 130 and a second electrode 140. The second electrode 140 of the anti-fuse structure AF is disposed adjacent to and spaced apart from the first electrode 130 of the anti-fuse structure AF. Both of a top surface 131 of the first electrode 130 of the anti-fuse structure AF and a top surface 141 of the second electrode 140 of the anti-fuse structure AF are below a top surface 111 of the semiconductor substrate 110. Since an entirety of the anti-fuse structure AF is disposed within the isolation structure 120 (or the semiconductor substrate 110), a size of the memory device 10 can be decreased. Further, a manufacturing process of the memory device 10 can be simplified.

The memory device 10 further includes a first contact 150 and a second contact 160 adjacent to the first contact 150. The first contact 150 is disposed on the first electrode 130 of the anti-fuse structure AF, and the second contact 160 is

disposed on the second electrode 140 of the anti-fuse structure AF. In greater details, the first contact 150 is in contact with and electrically connected to the first electrode 130 and the second contact 160 is in contact with and electrically connected to the second electrode 140.

Programming mechanism by using the anti-fuse structure AF to store digital information is to apply the first voltage V1 to the first electrode 130 through the first contact 150 and apply the second voltage V2 to the second electrode 140 through the second contact 160 to convert a portion 122 of the isolation structure 120 between the first electrode 130 and the second electrode 140 into a permanent electrically conductive path 122a so as to conduct between the first electrode 130 and the second electrode 140 of the anti-fuse structure AF, such that the anti-fuse structure AF is in an “On” state. Conversely, the unprogrammed anti-fuse structure AF is in an “Off” state. It is noted that a voltage difference (also referred as voltage bias) between the first voltage V1 and the second voltage V2 may exceed a certain level (e.g., exceed a predetermined voltage bias) to program the anti-fuse structure AF. In some embodiments, the voltage V1 connected to the first electrode 130 through the first contact 150 is higher than the voltage V2 connected to the second electrode 140 through the second contact 160. For example, the voltage V1 is a high voltage and the voltage V2 is a low voltage or at zero (ground) voltage. It is noted that the anti-fuse structure AF further includes a portion 122 of the isolation structure 120 between the first electrode 130 and the second electrode 140, and the portion 122 of the isolation structure 120 is referred as an anti-fuse dielectric material of the anti-fuse structure AF. The anti-fuse structure AF is programmed by applying a voltage bias across the first electrode 130 and the second electrode 140 to rupture the portion 122 of the isolation structure 120 (i.e., anti-fuse dielectric material) and form the permanent electrically conductive path 122a between the first electrode 130 and the second electrode 140. That is, the portion 122 of the isolation structure 120 is changed from non-conductive (i.e., dielectric material) to conductive (i.e., conductor) when programming the anti-fuse structure AF.

In some embodiments, both of the top surface 131 of the first electrode 130 of the anti-fuse structure AF and the top surface 141 of the second electrode 140 of the anti-fuse structure AF are below a top surface 121 of the isolation structure 120. In other words, an entirety of the anti-fuse structure AF (including the first electrode 130, the second electrode 140 and the portion 122 of the isolation structure 120 between the first electrode 130 and the second electrode 140) is within the isolation structure 120 (or the semiconductor substrate 110).

In some embodiments, a height H1 of the first electrode 130 of the anti-fuse structure AF is substantially the same as a height H2 of the second electrode 140 of the anti-fuse structure AF. In greater details, the top surface 131 of the first electrode 130 of the anti-fuse structure AF is substantially coplanar with the top surface 141 of the second electrode 140 of the anti-fuse structure AF, and a bottom surface 133 of the first electrode 130 of the anti-fuse structure AF is substantially coplanar with a bottom surface 143 of the second electrode 140 of the anti-fuse structure AF. In some embodiments, the height H1 of the first electrode 130 of the anti-fuse structure AF is in a range of about 130 nanometers (nm) to about 150 nanometers (e.g., 140 nanometers), and the height H2 of the second electrode 140 of the anti-fuse structure AF is in a range of about 130 nanometers to about 150 nanometers (e.g., 140 nanometers). As such, the

first electrode **130** and the second electrode **140** of the anti-fuse structure AF can have better uniformity.

In some embodiments, a thickness T1 of the first electrode **130** of the anti-fuse structure AF is substantially the same as a thickness T2 of the second electrode **140** of the anti-fuse structure AF. The thickness T1 of the first electrode **130** of the anti-fuse structure AF is in a range of about 70 nanometers to about 90 nanometers (e.g., 80 nanometers), and the thickness T2 of the second electrode **140** of the anti-fuse structure AF is in a range of about 70 nanometers to about 90 nanometers (e.g., 80 nanometers). As such, the permanent electrically conductive path **122a** between the first electrode **130** and the second electrode **140** of the anti-fuse structure AF can be easily formed during programming.

In some embodiments, a distance D1 between the first electrode **130** of the anti-fuse structure AF and the second electrode **140** of the anti-fuse structure AF (i.e., a length of the portion **122** of the isolation structure **120** from the first electrode **130** to the second electrode **140**) is in a range of about 20 nanometers to about 35 nanometers (e.g., 27 nanometers). As such, the permanent electrically conductive path **122a** can be formed between the first electrode **130** and the second electrode **140**. Further, the size of the anti-fuse structure AF can be decreased. If the distance D1 between the first electrode **130** and the second electrode **140** is less than about 20 nanometers, an interference would occur, thereby adversely affecting performance of the memory device **10**; if the distance D1 between the first electrode **130** and the second electrode **140** is greater than about 35 nanometers, the size of the anti-fuse structure AF (or memory device **10**) would be too large. In some embodiments, a size of the anti-fuse structure AF is about 240-250 nanometers (e.g., 248 nanometers) multiple about 240-250 nanometers (e.g., 248 nanometers). With such configuration of the anti-fuse structure AF discussed above, the size of the anti-fuse structure AF can be decreased. For example, the size of the anti-fuse structure AF can be decreased about 12%.

In some embodiments, the first electrode **130** of the anti-fuse structure AF and the second electrode **140** of the anti-fuse structure AF have the same profiles, such as tapered profiles. For example, the first electrode **130** of the anti-fuse structure AF includes a first bottom portion and a first top portion wider than the first bottom portion, and the second electrode **140** of the anti-fuse structure AF includes a second bottom portion and a second top portion wider than the second bottom portion.

The memory device **10** further includes a dielectric layer **170** above the semiconductor substrate **110** and the isolation structure **120**. The dielectric layer **170** surrounds the first contact **150** and the second contact **160**, and the isolation structure **120** surrounds the first electrode **130** of the anti-fuse structure AF and the second electrode **140** of the anti-fuse structure AF. In some embodiments, the first contact **150** includes a bottom portion **152** in the isolation structure **120** and a top portion **154** in the dielectric layer **170**. The second contact **160** includes a bottom portion **162** in the isolation structure **120** and a top portion **164** in the dielectric layer **170**. In some embodiments, both of a bottom surface **153** of the first contact **150** and a bottom surface **163** of the second contact **160** are below the top surface **111** of the semiconductor substrate **110**. The bottom surface **153** of the first contact **150** is spaced apart from the dielectric layer **170** and the bottom surface **163** of the second contact **160** is spaced apart from the dielectric layer **170** as well.

In some embodiments, the bottom portion **152** of the first contact **150** and the top portion **154** of the first contact **150**

have different profiles. For example, the bottom portion **152** of the first contact **150** has a tapered profile and the top portion **154** of the first contact **150** has a rectangle profile. The bottom portion **152** of the first contact **150** has a minimum width smaller than that of the top portion **154** of the first contact **150**. Similarly, the bottom portion **162** of the second contact **160** and the top portion **164** of the second contact **160** have different profiles. For example, the bottom portion **162** of the second contact **160** has a tapered profile and the top portion **164** of the second contact **160** has a rectangle profile. The bottom portion **162** of the second contact **160** has a minimum width smaller than that of the top portion **164** of the second contact **160**.

In some embodiments, as shown in FIG. 1 (top view), the first electrode **130** of the anti-fuse structure AF extends along a first direction D1, and the second electrode **140** of the anti-fuse structure AF extends along the first direction D1. The first electrode **130** of the anti-fuse structure AF and the second electrode **140** of the anti-fuse structure AF are arranged parallel with each other along a second direction D2 perpendicular to the first direction D1. In some embodiments, the first electrode **130** of the anti-fuse structure AF and the second electrode **140** of the anti-fuse structure AF have the same profiles in the top view. For example, the first electrode **130** of the anti-fuse structure AF and the second electrode **140** of the anti-fuse structure AF have strip profiles.

In some embodiments, as shown in FIG. 1, the first contact **150** is disposed above an end of the first electrode **130** and the second contact **160** is disposed above an end of the second electrode **140**, wherein the end of the first electrode **130** is misaligned with the end of the second electrode **140**. As such, the first contact **150** and the second contact **160** are misaligned with each other. If the first contact **150** is aligned with the second contact **160** (e.g., the first contact **150** and the second contact **160** are arranged in the second direction D2), the first contact **150** would be too close to the second contact **160**, thereby causing interference thereof.

In some embodiments, the first electrode **130** of the anti-fuse structure AF and the second electrode **140** of the anti-fuse structure AF include the same materials. For example, the first electrode **130** of the anti-fuse structure AF and the second electrode **140** of the anti-fuse structure AF include metal (e.g., titanium), metal nitride (e.g., titanium nitride), or other suitable conductive materials. The first contact **150** and the second contact **160** may include the same material, such as tungsten, or other suitable metal materials.

FIG. 3 to FIG. 6 are cross-sectional views of a method of forming the memory device at various stages in accordance with some embodiments of the present disclosure.

Referring to FIG. 3, the semiconductor substrate **110** is provided. In some embodiments, the semiconductor substrate **110** includes silicon. In some other embodiments, the semiconductor substrate **110** includes another elementary semiconductor, such as germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

The isolation structure **120** is formed in the semiconductor substrate **110**. In some embodiments, the isolation structure **120** is shallow trench isolation (STI). The formation of the isolation structure **120** may include etching the semiconductor substrate **110** to form a trench **124** in the semi-

7

conductor substrate **110** and then filling an anti-fuse dielectric material in the trench **124**. The anti-fuse dielectric material may include insulator materials such as silicon dioxide. In some embodiments, the isolation structure **120** is formed by physical vapor deposition (PVD), chemical vapor deposition (CVD), or the like.

Referring to FIG. 4, after the isolation structure **120** is formed, an etching process is performed to form a first opening **O1** and a second opening **O2** in the isolation structure **120** such that the isolation structure **120** is partially removed. The opening **O1** and the second opening **O2** expose underlying isolation structure **120**.

Referring to FIG. 4 and FIG. 5, the first electrode **130** and the second electrode **140** of the anti-fuse structure AF are respectively formed in the first opening **O1** and the second opening **O2** such that both of the top surface **131** of the first electrode **130** of the anti-fuse structure AF and the top surface **141** of the second electrode **140** of the anti-fuse structure AF are below the top surface **121** of the isolation structure **120**. Further, both of the top surface **131** of the first electrode **130** of the anti-fuse structure AF and the top surface **141** of the second electrode **140** of the anti-fuse structure AF are below the top surface **111** of the semiconductor substrate **110**.

In some embodiments, forming the first electrode **130** and the second electrode **140** of the anti-fuse structure AF includes filling conductive materials in the first opening **O1** and the second opening **O2** to respectively form a first electrode layer and a second electrode layer in the isolation structure **120**, and then etching back the first electrode layer and the second electrode layer to respectively form the first electrode **130** and the second electrode **140**.

In some embodiments, each of the first electrode **130** and the second electrode **140** is a single layer and includes, for example, titanium (Ti) or titanium nitride (TiN). In some embodiments, both of the first electrode **130** and the second electrode **140** are a multi-layered structure and include, for example, a titanium nitride layer and a tungsten layer above the titanium nitride layer. Formation of the first electrode **130** and the second electrode **140** may be exemplarily performed using a CVD process, a PVD process, an ALD process, the like, and/or a combination thereof.

In some embodiments, the first electrode **130** and the second electrode **140** of the anti-fuse structure AF are formed such that the portion **122** of the isolation structure **120** is directly between the first electrode **130** and the second electrode **140**, and the portion **122** of the isolation structure is referred as the anti-fuse dielectric material of the anti-fuse structure AF. In some embodiments, the first electrode **130** and the second electrode **140** of the anti-fuse structure AF are surrounded by and in contact with the isolation structure **120**.

After the first electrode **130** and the second electrode **140** of the anti-fuse structure AF is formed, a dielectric layer **170** is formed above the isolation structure **120** and the semiconductor substrate **110**. The dielectric layer **170** may include a first portion **172** and a second portion **174** within the isolation structure, wherein the first portion **172** is in contact with the first electrode **130** and the second portion **174** is in contact with the second electrode **140**. The dielectric layer **170** has a bottommost surface below the top surface **121** of the isolation structure **120** (or the top surface **111** of the semiconductor substrate **110**). The dielectric layer **170** may be formed by CVD, high-density plasma CVD, spin-on, sputtering, or other suitable methods. In some embodiments, the dielectric layer **170** is formed of dielectric material different from the isolation structure. For example,

8

the dielectric layer **170** includes nitride (e.g., silicon nitride), while the isolation structure **120** includes oxide (e.g., silicon dioxide).

Referring to FIGS. 5 and 6, after the dielectric layer **170** is formed, an etching process is performed to form a third opening **O3** and a fourth opening **O4** in the dielectric layer such that the dielectric layer **170** is partially removed. The etching process removes the first portion **172** and the second portion **174** of the dielectric layer **170** such that the first electrode **130** and the second electrode **140** of the anti-fuse structure AF are exposed.

Referring back to FIG. 2, after the third opening **O3** and the fourth opening **O4** (see FIG. 6) are formed, the first contact **150** and the second contact **160** are respectively formed in the third opening **O3** and the fourth opening **O4** (see FIG. 6). The first voltage **V1** is applied to the first electrode **130** of the anti-fuse structure AF through the first contact **150** and the second voltage different from the first voltage is applied to the second electrode **140** of the anti-fuse structure through the second contact **160** to convert the portion **122** of the isolation structure **120** between the first electrode **130** and the second electrode **140** of the anti-fuse structure AF into a permanent electrically conductive path. As a result, the memory device **10** as shown in FIG. 2 can be obtained.

Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

What is claimed is:

1. A memory device, comprising:

a semiconductor substrate;
an isolation structure in the semiconductor substrate; and
an anti-fuse structure in the isolation structure, the anti-fuse structure comprising:

a first electrode; and
a second electrode adjacent to the first electrode, wherein both of a top surface of the first electrode and a top surface of the second electrode are below a top surface of the semiconductor substrate, wherein both of the top surface of the first electrode and the top surface of the second electrode are below a top surface of the isolation structure.

2. The memory device of claim 1, further comprising:
a first contact on the first electrode; and
a second contact on the second electrode.

3. The memory device of claim 2, wherein both of a bottom surface of the first contact and a bottom surface of the second contact are below the top surface of the semiconductor substrate.

4. The memory device of claim 2, further comprising:
a dielectric layer above the semiconductor substrate and the isolation structure.

5. The memory device of claim 4, wherein the first contact includes a bottom portion in the isolation structure and a top portion in the dielectric layer.

6. The memory device of claim 2, wherein the first contact is configured to apply a first voltage to the first electrode and the second contact is configured to apply a second voltage

9

different from the first voltage to the second electrode to convert a portion of the isolation structure between the first electrode and the second electrode into a permanent electrically conductive path.

7. The memory device of claim 1, wherein the top surface of the first electrode is substantially coplanar with the top surface of the second electrode. 5

8. The memory device of claim 1, wherein a bottom surface of the first electrode is substantially coplanar with a bottom surface of the second electrode. 10

9. The memory device of claim 1, wherein the first electrode extends along a first direction, and the first electrode and the second electrode are arranged parallel with each other along a second direction perpendicular to the first direction in a top view. 15

10. The memory device of claim 1, wherein the first electrode has a strip profile in a top view. 20

11. The memory device of claim 1, wherein the first electrode and the second electrode comprise the same materials. 25

12. A method of forming memory device, comprising:
forming an isolation structure in a semiconductor substrate;

etching the isolation structure to form a first opening and a second opening; 25

forming a first electrode and a second electrode of an anti-fuse structure respectively in the first opening and the second opening such that both of a top surface of the first electrode and a top surface of the second electrode are below a top surface of the isolation structure; 30

forming a dielectric layer above the isolation structure; and

applying a first voltage to the first electrode and a second voltage different from the first voltage to the second electrode to convert a portion of the isolation structure 35

10

between the first electrode and the second electrode into a permanent electrically conductive path.

13. The method of claim 12, further comprising:
forming a first contact and a second contact in the dielectric layer such that the first voltage is applied to the first electrode through the first contact and the second voltage is applied to the second electrode through the second contact.

14. The method of claim 13, wherein forming the first contact and the second contact is performed such that a portion of the first contact and a portion of the second contact are formed within the isolation structure. 10

15. The method of claim 13, wherein forming the first contact and the second contact is performed such that the top surface of the first electrode and the top surface of the second electrode are respectively in contact with the first contact and the second contact. 15

16. The method of claim 12, wherein forming the first electrode and the second electrode is performed further such that the top surface of the first electrode and the top surface of the second electrode are below a top surface of the semiconductor substrate. 20

17. The method of claim 12, wherein forming the first electrode and the second electrode is performed such that the first electrode and the second electrode are in contact with the isolation structure. 25

18. The method of claim 12, wherein forming the first electrode and the second electrode is performed such that a bottom surface of the first electrode is coplanar with a bottom surface of the second electrode. 30

19. The method of claim 12, wherein forming the isolation structure in the semiconductor substrate comprises:
forming a trench in the semiconductor substrate; and
filling an anti-fuse dielectric material in the trench to form the isolation structure. 35

* * * * *